VOLTWISE: SYSTEM FEATURE AUDIT & MANUAL CHECKLIST

Comprehensive Hardware, Firmware, Machine Learning, Backend API & Mobile Application Verification Document
Purpose: Print-ready field audit protocol for hardware bench testing, software validation, and final project defense.

Target Workspace:	voltwise - dev (Monorepo)	**Inspection Date:**	____ / ____ / 2026
System Version:	v1.0.0 (MVP Release)	**Lead Inspector:**	_____
Target ESP32 UID:	esp32-01 (or custom UID)	**Affiliation / Role:**	_____

System Completion Scorecard & Manual Testing Instructions

Subsystem / Architectural Tier	Total	Completed	Partial	Unfinished	Completion Rate
1. IoT Hardware & ESP32 Firmware	14	11	1	2	**78.6%**
2. Machine Learning / NILM Service	12	10	1	1	**83.3%**
3. Backend REST & Ingestion API	18	15	2	1	**83.3%**
4. Database & Telemetry Storage	8	6	2	0	**75.0%**
5. Mobile Application (Frontend)	22	19	0	3	**86.4%**
6. Cross-Cutting & Security	8	7	1	0	**87.5%**
TOTAL SYSTEM SCORE	**82**	**68**	**7**	**7**	**82.9% MVP Ready**

HOW TO USE THIS MANUAL CHECKLIST: When performing physical bench testing or evaluating the app, use a pen to check: **[P]** = **PASS** (Feature physically tested and confirmed functioning as specified). **[F]** = **FAIL** (Defect, malfunction, or crash observed). Use the **Field Notes & Defect Log** column to document exact measurements, multimeter readings, or error messages.

1. IoT Hardware & Firmware Checklist (iot-voltwise/)

Check	Feature & Verification Criteria	Status	Code / Pin	Physical Verification Notes

[ ] P
[ ] F
	PZEM-004T UART Telemetry
	Reads V, I, W, kWh, Hz, PF every 2s over Hardware Serial2. Multimeter check: 220V AC, verify non-zero W on live load.
		PASS (Ready)	`GPIO 16 (RX2)`
			`GPIO 17 (TX2)`
			`main.cpp#L25`
				[ ] AC Volts: _____ V
				[ ] Current: _____ A

[ ] P [ ] F	**PZEM Diagnostic Boot Ping** ESP32 prints slave address and raw UART packet bytes on boot serial (115200 baud). Differentiates UART fault from dead mains.	**PASS (Ready)**	`main.cpp#L69` `DIAGNOSTIC=true`	[ ] UART Packet OK [ ] Slave Addr verified

[ ] P
[ ] F
	Dual-Tier Relay Control
	Pin 25 switches Low Load (10A module); Pin 26 switches High Load (30A module). Active-LOW logic.
		PASS (Ready)	`Pin 25 (Low)`
			`Pin 26 (High)`
			`main.cpp#L30`
				[ ] Low Relay click heard
				[ ] High Relay click heard

[ ] P [ ] F	**Staggered Power Sequencing** When turning ON master/all, Low Line energizes immediately; High Line energizes after 3s delay to prevent inrush brownout.	**PASS (Ready)**	`INTERVAL_TIME=3` `main.cpp#L50`	[ ] 3-second delay timed [ ] No ESP32 reset observed
[ ] P [ ] F	**Smart Hybrid Boot Power** Searches Wi-Fi for 15s. If no router connects, falls back to Autonomous Mode and powers up relays so house is never left dark.	**PASS (Ready)**	`TIMEOUT_SEC=15` `main.cpp#L57`	[ ] Router OFF test: power energized after 15s timeout

Check	Feature & Verification Criteria	Status	Code / Pin	Physical Verification Notes
[ ] P [ ] F	**Dynamic Over-Power Cutoff** Trips relays immediately when power exceeds threshold (default 3000W). Can be reconfigured in real-time over MQTT.	PASS (Ready)	`voltwise/<uid>/ safety/set`	[ ] Tested cutoff at threshold [ ] Relays latched OFF
[ ] P [ ] F	**Auto-Shutdown Countdown** Starts countdown when load > 0.05A is sensed; cuts power when expired (default 30 min, settable via MQTT).	PASS (Ready)	`voltwise/<uid>/ countdown/set`	[ ] Countdown decrementing [ ] Automatic cutoff tested
[ ] P [ ] F	**SoftAP Captive Portal Server** Hosts 'VoltWise-Setup-XXXX' AP. Mobile phone connection prompts captive portal login web page at 192.168.4.1.	PASS (Ready)	`provisioning_ server.h#L1`	[ ] AP SSID visible on phone [ ] Web portal rendered OK
[ ] P [ ] F	**Wi-Fi Scan & Provision API** Endpoints /api/scan and /api/provision allow scanning nearby 2.4GHz SSIDs and saving credentials into flash.	PASS (Ready)	`provisioning_ server.h#L52`	[ ] Scan returns home SSID [ ] Saves pass & connects
[ ] P [ ] F	**NVS Credential Persistence** ESP32 Preferences library stores SSID, password, and custom UID in flash. Survives full power cycle.	PASS (Ready)	`config_manager.h prefs.begin()`	[ ] Power cycle test passed [ ] Auto-reconnects to Wi-Fi
[ ] P [ ] F	**Hardware Factory Reset Button** Pressing & holding ESP32 BOOT button (GPIO 0) for 5 seconds wipes flash credentials and re-launches Setup AP.	PASS (Ready)	`GPIO 0 (BOOT) main.cpp#L279`	[ ] Hold 5s confirmed [ ] Flash wiped, AP restarted
[ ] P [ ] F	**TLS HiveMQ Cloud MQTT & LWT** Encrypted TLS on port 8883. Publishes telemetry every 2s. Publishes LWT 'offline' when board loses power.	PASS (Ready)	`port 8883 (TLS) main.cpp#L1316`	[ ] Broker connected [ ] LWT verified on disconnect
[ ] P [ ] F	**6 Individual Socket Switching** Architectural spec calls for 6 individual relays (Pins 12-15 + 25-26). Current firmware groups into 2 lines (Pins 25 & 26).	PARTIAL (2-Line)	`Schematic v4 vs main.cpp#L30`	[ ] Low Load Line (Pin 25) [ ] High Load Line (Pin 26)
[ ] P [ ] F	**OTA Wireless Firmware Flash** Over-The-Air flashing via ArduinoOTA or local web server without USB cable.	MISSING (USB Only)	`iot-voltwise/src/`	[ ] Must connect USB-C to flash via PlatformIO
[ ] P [ ] F	**Offline Flash Telemetry Spool** SPIFFS/LittleFS spooling of sensor readings when Wi-Fi drops, with replay on reconnection.	MISSING (Discarded)	`iot-voltwise/src/`	[ ] Readings lost if router offline during recording

2. Machine Learning / NILM Service Checklist (ml-voltwise/)

Check	Feature & Verification Criteria	Status	Code / Route	Physical Verification Notes
[ ] P [ ] F	**FastAPI Microservice Engine** Asynchronous Python service running Uvicorn on port 8000. GET /health returns model_loaded=True, best_k=7.	PASS (Ready)	`GET /health main.py#L204`	[ ] HTTP 200 OK received [ ] Model loaded in memory
[ ] P [ ] F	**30-Second Window Feature Extraction** Extracts 10 electrical features (V mean, I mean/max, P mean/max/min/std, Energy Wh, Freq, PF) from raw telemetry.	PASS (Ready)	`feature_ engineering.py`	[ ] 10 features computed [ ] Inrush spikes captured

Check	Feature & Verification Criteria	Status	Code / Route	Physical Verification Notes
[ ] P [ ] F	**KMeans Clustering Pipeline** StandardScaler + KMeans with silhouette score optimization. Model artifact kmeans_pipeline.joblib trained & verified.	PASS (Ready)	`artifacts/ kmeans_pipeline`	[ ] Silhouette score: 0.5301 [ ] K=7 clusters loaded
[ ] P [ ] F	**Learned Appliance Signatures** Trained on 918 real windows (Aircon, Refrigerator, Light Bulb, Cellphone Charger Samsung A14, Electric Kettle).	PASS (Ready)	`cluster_profiles .csv`	[ ] AC: ~630W run, 3.6kW surge [ ] Fridge: ~27W run, 190W comp
[ ] P [ ] F	**Euclidean Anomaly Detection** Flags electrical windows exceeding 95th percentile distance threshold (1.5352) to nearest centroid as 'warning' anomalies.	PASS (Ready)	`threshold=1.5352 main.py#L160`	[ ] High-distance load flagged [ ] Anomaly warning generated
[ ] P [ ] F	**Live Telemetry Stream Buffer** POST /telemetry ingests 2s readings into sliding time window per panel; returns prediction once >= 3 readings arrive.	PASS (Ready)	`POST /telemetry main.py#L244`	[ ] Pruning older than 30s [ ] Status transitions to predicted
[ ] P [ ] F	**Batch / Vector Predict API** POST /predict accepts direct 10-feature vector dictionary for instantaneous classification and offline analysis.	PASS (Ready)	`POST /predict main.py#L229`	[ ] Vector inference verified [ ] test_api.ps1 script passes
[ ] P [ ] F	**On-Demand Model Retrain API** POST /train triggers re-clustering over all CSV files inside data/raw/ and auto-reloads the active pipeline.	PASS (Ready)	`POST /train main.py#L293`	[ ] Retrains on new CSV files [ ] Re-calculates best K
[ ] P [ ] F	**Backend Bridge & Device Matching** Node.js backend (mlService.ts) relays MQTT telemetry to ML service; maps dominant appliance to PostgreSQL Device table.	PASS (Ready)	`mlService.ts backend/src/lib`	[ ] Matched device -> ACTIVE [ ] Standby (<3W) -> baseline
[ ] P [ ] F	**Unregistered Load Auto-Alert** Unrecognized appliance triggers INFO alert: 'VoltWise ML detected a [Appliance] (~W). Do you want to register this device?'	PASS (Ready)	`matchOrTagDevice mlService.ts#L290`	[ ] Alert visible in app [ ] 2-min alert cooldown active
[ ] P [ ] F	**True Multi-Load Disaggregation** Disaggregating multiple concurrent overlapping appliances (e.g. Fridge + AC running simultaneously on same line).	PARTIAL (Dominant)	`KMeans model architecture`	[ ] Identifies dominant load [ ] Cannot separate simultaneous
[ ] P [ ] F	**Active Learning Retrain Loop** User registering an unregistered load automatically triggers retraining of the KMeans model with new labeled data.	MISSING (Manual)	`ml-voltwise/ backend`	[ ] Requires running train.py or POST /train manually

3. Backend REST & Ingestion API Checklist (backend/)

Check	Feature & Verification Criteria	Status	Endpoint / File	Physical Verification Notes
[ ] P [ ] F	**JWT Authentication & Passwords** POST /api/auth/register and /login issue 7-day JWTs. Passwords securely hashed with bcrypt (salt rounds 10).	PASS (156 Tests)	`POST /api/auth/ register, login`	[ ] Auth tokens validated [ ] Bcrypt hash verified in DB
[ ] P [ ] F	**Revocable Multi-Session Tracking** Each sign-in writes a Session row. Token embeds sid claim. RequireAuth validates session status on every protected request.	PASS (Ready)	`requireAuth sessions.ts`	[ ] Revoked token blocked (401) [ ] lastSeenAt refreshed 1/min

Check	Feature & Verification Criteria	Status	Endpoint / File	Physical Verification Notes
[ ] P [ ] F	**Security Session Audit & Revoke** GET /api/security/overview lists active devices. DELETE /sessions/:id revokes one; DELETE /sessions revokes other sessions.	PASS (Ready)	`/api/security/ overview, sessions`	[ ] Session list matches phones [ ] Remote sign-out tested
[ ] P [ ] F	**Account Cascade Deletion** DELETE /api/security/account requires password verification and cascades deletion of all rooms, devices, readings, and alerts.	PASS (Ready)	`DELETE /api/ security/account`	[ ] Password check enforced [ ] User data wiped cleanly
[ ] P [ ] F	**Virtual Device Inventory CRUD** GET, POST, PATCH, DELETE /api/devices. Manages device metadata, rated watts, and status (ACTIVE, IDLE, OFF, UNPOWERED).	PASS (Ready)	`/api/devices devices.service.ts`	[ ] Create & Edit tested [ ] Status transitions verified
[ ] P [ ] F	**Device Photo Uploads** POST /api/devices/:id/photo accepts multipart image (5MB max, JPEG/PNG/WebP), saves in backend/uploads/, served at /uploads.	PASS (Ready)	`POST /api/devices /:id/photo`	[ ] Image saved in uploads/ [ ] Photo renders in app
[ ] P [ ] F	**Master Power Reconciliation** POST /api/devices/reconcile-power re-evaluates all device statuses when master relay is toggled. Marks devices UNPOWERED.	PASS (Ready)	`POST /api/devices /reconcile-power`	[ ] Master OFF -> CIRCUIT OFF [ ] Master ON -> restores states
[ ] P [ ] F	**Live Dashboard Aggregation** GET /api/dashboard returns currentKw, totalTodayKwh, live reading snapshot, iotOnline, and bucketed Usage History.	PASS (Ready)	`GET /api/ dashboard`	[ ] Hero card is real-time [ ] Charts follow range filter
[ ] P [ ] F	**Unified Time-Range Engine** src/lib/range.ts handles Day (24h), Week (7d), Month (28-31d), and Cycle. Future time buckets strictly dropped.	PASS (Ready)	`src/lib/range.ts range.test.ts`	[ ] Manila time (UTC+8) OK [ ] No flat line into future
[ ] P [ ] F	**Analytics & Bill Predictor API** GET /api/analytics computes accumulated kWh, estimated monthly bill based on tariff, category breakdown, and top consumers.	PASS (Ready)	`GET /api/ analytics`	[ ] Bill projection calculated [ ] Min/avg/max stats verified
[ ] P [ ] F	**Tariff Rate Management** GET /api/analytics/tariff and PUT /api/analytics/tariff update user's rate per kWh (default 10.50 P) and currency symbol.	PASS (Ready)	`/api/analytics/ tariff`	[ ] Rate change saved to DB [ ] Future cost calculations sync
[ ] P [ ] F	**Alert Feed & Badge Tracking** GET /api/alerts, POST /read-all, PATCH /:id/read. Supports CRITICAL, WARNING, INFO alerts with unread counter.	PASS (Ready)	`/api/alerts alerts.service.ts`	[ ] Badge count decrements [ ] Mark all as read tested
[ ] P [ ] F	**New-Load Step Detector** Pure I/O-free EMA baseline detector (loadDetector.ts). Emits INFO alert when sustained power jump is observed.	PASS (Ready)	`src/lib/ loadDetector.ts`	[ ] Load detect test passes [ ] Non-blocking fire-and-forget
[ ] P [ ] F	**MQTT Ingest & Delta kWh Engine** Ingests 2s telemetry into EnergyReading. Computes exact incremental delta kWh to avoid raw PZEM odometer multiplying errors.	PASS (Ready)	`src/lib/mqtt.ts activeDeltaKwh`	[ ] SUM(kwh) stays accurate [ ] Ingests for demo user
[ ] P [ ] F	**Data Collection Controller** POST /collection/start, /stop, GET /status, DELETE /readings allow labeling real appliance telemetry and purging test data.	PASS (Ready)	`/api/iot/ collection`	[ ] Session tracking active [ ] Readings tagged to device

Check	Feature & Verification Criteria	Status	Endpoint / File	Physical Verification Notes
[ ] P [ ] F	**Data Export Engine (CSV/JSON)** GET /api/export/:dataset streams readings, devices, alerts, security logs with formula-injection guard (prepends ').	PASS (Ready)	`/api/export` `export.service.ts`	[ ] CSV opened in Excel OK [ ] Capped at 20,000 rows
[ ] P [ ] F	**Password Reset via Email** POST /api/auth/forgot-password returns resetToken directly in HTTP JSON body. No external SMTP provider configured.	PARTIAL (MVP Bypass)	`POST /api/auth/` `forgot-password`	[ ] Token returned in response [ ] No real SMTP email sent
[ ] P [ ] F	**Automated Billing Period Rollover** Automatic monthly cron job to close expired BillingPeriod rows and archive accumulated kWh snapshot in PostgreSQL.	MISSING (No Cron)	`backend/src/` `schema.prisma`	[ ] Client calculates cycle [ ] No server-side cron worker

4. Database & Telemetry Storage Checklist (PostgreSQL + Prisma)

Check	Feature & Verification Criteria	Status	Model / Migration	Physical Verification Notes
[ ] P [ ] F	**Prisma Relational Data Models** 7 models: User, Session, Tariff, BillingPeriod, Room, Device, EnergyReading, Alert. Enforces foreign key constraints.	PASS (Ready)	`prisma/` `schema.prisma`	[ ] Migrations applied OK [ ] Cascade rules active
[ ] P [ ] F	**Dual-Use Telemetry Design** EnergyReading stores both whole-home aggregate (deviceId: null) and per-appliance (deviceId: int) time-series telemetry.	PASS (Ready)	`EnergyReading` `schema.prisma#L132`	[ ] deviceId=null for panel [ ] deviceId=N for appliance
[ ] P [ ] F	**Time-Series Performance Indexes** Compound indexes on [deviceId, timestamp] and [userId, timestamp] on EnergyReading; [userId, lastSeenAt] on Session.	PASS (Ready)	`@@index([...])` `schema.prisma`	[ ] Fast range lookups [ ] Zero sequential table scans
[ ] P [ ] F	**Prisma 7 Postgres Adapter** Prisma client configured with @prisma/adapter-pg connection pool in src/lib/prisma.ts.	PASS (Ready)	`src/lib/prisma.ts` `@prisma/adapter-pg`	[ ] Connection pooling OK [ ] Reused across all modules
[ ] P [ ] F	**Comprehensive Demo Seed Data** npm run seed creates demo@voltwise.app / password123 with 5 rooms, 7 devices, 30 days of hourly readings, and alerts.	PASS (Ready)	`prisma/seed.ts` `demo@voltwise.app`	[ ] Seed executes cleanly [ ] Demo household populated
[ ] P [ ] F	**Device Photo Local Disk Bucket** Device photos stored on local disk in backend/uploads/. Static file serving configured at Express /uploads route.	PASS (Ready)	`backend/uploads/` `index.ts#L45`	[ ] Files saved with device id [ ] Statically accessible
[ ] P [ ] F	**TimescaleDB Continuous Aggregates** Hypertables with automated rollups of raw 2s telemetry into 1-minute, 1-hour, and 1-day continuous aggregate views.	PARTIAL (Standard PG)	`docs/ vs active` `schema.prisma`	[ ] Operates on standard PG [ ] Hypertables not configured
[ ] P [ ] F	**Cloud Blob Object Storage** S3 or Google Cloud Storage integration for scalable, resilient photo uploads with pre-signed CDN URLs.	PARTIAL (Local Disk)	`backend/uploads/` `upload.ts`	[ ] Local filesystem used [ ] Production requires S3

5. Mobile Application Frontend Checklist (app-voltwise/)

Check	Feature & Verification Criteria	Status	Screen / Component	Physical Verification Notes
[ ] P [ ] F	**Real-Time PZEM Telemetry Grid** Live expandable metrics grid (V, I, W, kWh, Hz, PF). Subscribes directly to HiveMQ over WSS. Stale warning if > 10s gap.	PASS (Ready)	`dashboard.tsx` `METRIC_DEFS`	[ ] Telemetry streaming live [ ] Grid expands & collapses
[ ] P [ ] F	**Multi-Line Usage History Chart** Whole-home line + per-device curves. 3-hour fixed axis ticks. Interactive touch tooltip identifies exact hour and kWh.	PASS (Ready)	`dashboard.tsx` `DeviceUsageChart`	[ ] Tap points show tooltip [ ] Axis labels clear & readable
[ ] P [ ] F	**Time-Range Selector & Date Picker** RangeNavigator chip control (Day, Week, Month, Cycle). Custom DatePickerModal allows choosing past calendar months.	PASS (Ready)	`RangeNavigator.tsx` `DatePickerModal`	[ ] Future arrow disabled [ ] Range switches correctly
[ ] P [ ] F	**Virtual Device Inventory Registry** Room filter chips, search bar, banner vs photo card view mode. Badges: ACTIVE, IDLE, OFF, CIRCUIT OFF.	PASS (Ready)	`devices.tsx` `STATUS_LABELS`	[ ] Search filters instantly [ ] Card photos render cleanly
[ ] P [ ] F	**Camera & Gallery Device Photos** Expo ImagePicker opens phone camera or photo gallery. Uploads image to backend with upload progress spinner.	PASS (Ready)	`devices.tsx` `expo-image-picker`	[ ] Camera snap succeeds [ ] Upload saves to device
[ ] P [ ] F	**Software Device Toggle & Master Lock** Appliance toggle with optimistic update. If master power is cut, toggle is disabled and informs user why.	PASS (Ready)	`devices.tsx` `MASTER_OFF_LABEL`	[ ] Toggle updates state [ ] Blocked when master power off
[ ] P [ ] F	**Unregistered Load Quick Register** When ML detects an unregistered appliance, a banner appears with a one-tap shortcut to save it into the user inventory.	PASS (Ready)	`devices.tsx` `alerts.tsx`	[ ] Banner prompts user [ ] Pre-fills name and watts
[ ] P [ ] F	**Alerts Feed & Unread Badge** Severity filter chips (All, Critical, Warning, Info). Bottom tab bar badge counter synchronizes via ALERTS_CHANGED_EVENT.	PASS (Ready)	`alerts.tsx` `_layout.tsx badge`	[ ] Tab badge matches unread [ ] Filter chips work correctly
[ ] P [ ] F	**Anomaly Modal & Emergency Trip** Tapping an anomaly opens AnomalyModal showing watts, threshold, and recommendation, with an Emergency Cutoff button.	PASS (Ready)	`AnomalyModal.tsx` `alerts.tsx`	[ ] Modal details complete [ ] Emergency Cutoff trips relay
[ ] P [ ] F	**Center Power Control Tab Modal** Center tab (power.tsx) intercepts tap to open PowerControlModal. Controls Master Power, Line 1 (30A), Line 2 (10A).	PASS (Ready)	`power.tsx` `PowerControlModal`	[ ] Master toggle works [ ] High & Low Lines independent
[ ] P [ ] F	**Inrush Reconnection Safety Notice** Power modal displays safety banner explaining the 3-second staggered delay to protect against breaker tripping.	PASS (Ready)	`PowerControlModal` `#L712`	[ ] Safety banner visible [ ] Informs user during switch
[ ] P [ ] F	**Bill Predictor Cost Calculator** Accumulated kWh * user tariff rate = projected monthly bill. Uses daily burn rate across active billing cycle.	PASS (Ready)	`analytics.tsx` `formatCostOf`	[ ] Projected cost calculated [ ] Displays localized currency (P)
[ ] P [ ] F	**Consumption Donut Breakdown** SVG circular donut chart showing percentage consumption by appliance category with synchronized legend.	PASS (Ready)	`analytics.tsx` `getDonutSegments`	[ ] Donut slices render OK [ ] Percentages sum to 100%

Check	Feature & Verification Criteria	Status	Screen / Component	Physical Verification Notes
[ ] P [ ] F	**PZEM Electrical Health Stats** Accordion with Min/Avg/Max for Voltage, Current, Watts, Freq, PF with Philippine electrical safety standards context.	PASS (Ready)	`analytics.tsx` `FALLBACK_METRICS`	[ ] Voltage 220V +/-10% cited [ ] Accordion opens & closes
[ ] P [ ] F	**ESP32 Wi-Fi Provisioning Wizard** 5-step wizard (iot-provision.tsx): Hotspot connect -> Wi-Fi scan -> Password input -> ESP32 transfer -> Verification.	PASS (Ready)	`iot-provision.tsx` `app/iot-provision`	[ ] Scans 2.4GHz networks [ ] Connects ESP32 to router
[ ] P [ ] F	**Triple-Signal Link Diagnostics** Sensor & IoT screen checks app socket, backend connection, and ESP32 status. Pair to any UID without rebuilding app.	PASS (Ready)	`iot-settings.tsx` `linkHealth()`	[ ] Triple status indicators [ ] UID re-pairing succeeds
[ ] P [ ] F	**Biometric / Passcode App Lock** Expo LocalAuthentication gates signed-in app behind fingerprint / Face ID or device PIN. Provides sign-out escape hatch.	PASS (Ready)	`AppLockContext.tsx` `AppLockOverlay`	[ ] Fingerprint unlock works [ ] Escape hatch signs out cleanly
[ ] P [ ] F	**Appliance Data Collection Screen** Screen at /data-collection records PZEM telemetry for new appliances with session recovery and CSV export for ML.	PASS (Ready)	`data-collection.tsx` `app/data-collection`	[ ] Records telemetry to CSV [ ] Session survives app kill
[ ] P [ ] F	**Offline Sample Data Mode (DemoFab)** Floating flask button toggles realistic 7-device mock household data offline without API writes. Generates anomaly alerts.	PASS (Ready)	`DemoFab.tsx` `DemoDataContext`	[ ] Demo toggle populates UI [ ] In-memory only; no API writes
[ ] P [ ] F	**Dark / Light / System Theme Engine** Full theme toggle in settings with high-contrast, WCAG-compliant color palette. Persisted device-locally.	PASS (Ready)	`ThemeContext.tsx` `useTheme()`	[ ] Dark mode verified [ ] Light mode verified
[ ] P [ ] F	**Background Remote Push (APNs/FCM)** Push notifications when app is closed or killed. Currently runs locally in JS while app process is alive.	MISSING (In-App Only)	`NotificationContext.tsx#L360`	[ ] In-app banner works [ ] Background push not wired
[ ] P [ ] F	**Geofencing Automation Rules** Automation rules to automatically trip relays when user phone departs home GPS boundary.	MISSING (Planned)	`app-voltwise/app/`	[ ] No background location automation implemented

6. Hardware Bench Inspection & Observed Defect Log

Use this section during physical lab testing to record multimeter readings, oscilloscope checks, and anomalies.

Item #	Subsystem / Component	Observed Behavior / Measurement	Severity	Corrective Action / Resolution
1				
2				
3				
4				
5				

7. Inspection Sign-Off & Project Defense Certification

By signing below, the inspectors certify that the features enumerated in this audit document were physically verified against running hardware and codebase test suites, and that the recorded status reflects the true operational state of the VoltWise system.

LEAD DEVELOPER / PROPONENT:	HARDWARE / EMBEDDED ENGINEER:	TECHNICAL ADVISER / PANEL CHAIR:
_____ Signature over Printed Name Date: _____	_____ Signature over Printed Name Date: _____	_____ Signature over Printed Name Date: _____